

01. Introduction

Breast cancer is one of the most common cancers affecting women worldwide, although men can also develop it. It occurs when abnormal cells in the breast grow uncontrollably and form a tumor.

Early detection greatly increases the chances of successful treatment and long-term survival.

02. What is Breast Cancer?

Breast cancer develops when cells in the breast begin to divide uncontrollably.

The cancer may start in:

- Milk ducts (Ductal carcinoma)
- Milk-producing lobules (Lobular carcinoma)
- Connective tissues of the breast

If left untreated, breast cancer may spread to nearby lymph nodes and distant organs such as the lungs, liver, bones, and brain.

03. Risk Factors

Several factors increase the risk of breast cancer.

Non-Modifiable Risk Factors

- Increasing age
- Female gender
- Family history of breast cancer
- Genetic mutations (BRCA1 and BRCA2)
- Early menstruation
- Late menopause
- Dense breast tissue

Modifiable Risk Factors

- Obesity
 - Physical inactivity
 - Alcohol consumption
 - Smoking
 - Hormone replacement therapy
 - Poor diet
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04. Common Symptoms

Breast cancer may not cause symptoms in its early stages.

Common warning signs include:

- Lump in the breast
- Lump under the arm
- Breast swelling
- Change in breast shape
- Skin dimpling
- Nipple discharge
- Inverted nipple
- Redness of the breast
- Persistent breast pain
- Thickening of breast tissue

Any unusual breast change should be evaluated by a healthcare professional.

05. Breast Self-Examination (BSE)

Women are encouraged to become familiar with the normal appearance and feel of their breasts.

Self-examination may help identify:

- New lumps
- Skin changes
- Nipple abnormalities
- Breast swelling

Breast self-examination should not replace professional screening.

06. Clinical Breast Examination

A healthcare provider examines the breasts and nearby lymph nodes for abnormalities.

Clinical examinations are especially important for women at higher risk.

07. Mammography

Mammography is the most effective screening test for early breast cancer detection.

It can detect tumors before they become large enough to be felt.

Regular mammography significantly reduces breast cancer mortality.

08. Breast Ultrasound

Ultrasound uses sound waves to examine breast tissue.

It is commonly used:

- **To evaluate breast lumps**
 - **For women with dense breast tissue**
 - **Along with mammography**
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09. Breast MRI

Magnetic Resonance Imaging (MRI) provides highly detailed breast images.

MRI is recommended for:

- **High-risk women**
 - **BRCA mutation carriers**
 - **Complex breast abnormalities**
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10. Breast Biopsy

A biopsy confirms the diagnosis of breast cancer.

Types include:

- **Fine Needle Aspiration (FNA)**
- **Core Needle Biopsy**
- **Surgical Biopsy**

The tissue sample is examined under a microscope.

11. Breast Cancer Staging

Breast cancer is staged from Stage 0 to Stage IV.

Stage 0

Cancer remains within the milk ducts.

Stage I

Small tumor confined to the breast.

Stage II

Larger tumor or limited lymph node involvement.

Stage III

Cancer has spread to nearby lymph nodes or surrounding tissues.

Stage IV

Cancer has spread to distant organs.

12. Treatment Options

Treatment depends on:

- **Cancer stage**
 - **Tumor size**
 - **Hormone receptor status**
 - **HER2 status**
 - **Patient's age**
 - **Overall health**
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13. Surgery

Surgical options include:

Lumpectomy

Only the tumor and a small amount of surrounding tissue are removed.

Mastectomy

The entire breast is removed.

Some patients may also require lymph node removal.

14. Chemotherapy

Chemotherapy destroys rapidly growing cancer cells.

It may be given:

- **Before surgery (Neoadjuvant)**
- **After surgery (Adjuvant)**
- **For advanced breast cancer**

Common side effects include:

- **Hair loss**

- Fatigue
 - Nausea
 - Low blood counts
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15. Radiation Therapy

Radiation therapy destroys remaining cancer cells after surgery.

Benefits include:

- Lower recurrence risk
 - Improved survival
 - Better local disease control
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16. Hormone Therapy

Some breast cancers depend on estrogen or progesterone.

Hormone therapy blocks these hormones.

Common medicines include:

- Tamoxifen
- Aromatase inhibitors

Hormone therapy is often continued for several years.

17. Targeted Therapy

Targeted therapy attacks specific proteins involved in cancer growth.

Patients with HER2-positive breast cancer may receive HER2-targeted medicines.

Targeted therapy improves survival while reducing damage to normal cells.

18. Immunotherapy

Immunotherapy helps the immune system recognize and attack cancer cells.

It is mainly used in selected cases of advanced or triple-negative breast cancer.

19. Possible Complications

Untreated breast cancer may spread to:

- Bones

- Lungs
- Liver
- Brain

Early diagnosis greatly reduces these complications.

20. Prevention

Although breast cancer cannot always be prevented, risk can be reduced by:

- Maintaining a healthy weight
 - Regular physical activity
 - Limiting alcohol consumption
 - Avoiding smoking
 - Eating a balanced diet
 - Breastfeeding when possible
 - Participating in regular screening
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21. Follow-Up Care

After treatment, patients should:

- Attend regular follow-up appointments
- Perform breast self-awareness
- Undergo recommended imaging tests
- Report new symptoms immediately
- Maintain a healthy lifestyle

Regular follow-up helps detect recurrence early.

22. Key Points

- Breast cancer is one of the most common cancers worldwide.
- Early detection greatly improves survival.
- Mammography is the standard screening method.
- Diagnosis is confirmed through biopsy.
- Treatment includes surgery, chemotherapy, radiation therapy, hormone therapy, targeted therapy, and immunotherapy.

- **Healthy lifestyle choices and regular screening reduce risk.**
- **Long-term follow-up is essential after treatment.**